

One Attack to Fool Them All: Highly Transferable Black-Box Adversarial Attacks on Frontier MLLMs

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Abstract—Adversarial attacks have long posed a fundamental threat to machine learning systems. As multimodal large language models (MLLMs) rapidly evolve and become widely deployed, assessing their vulnerability to such attacks is essential for their safe use. In this work, we investigate whether a single adversarial image can consistently mislead diverse frontier MLLMs in black-box settings. We propose O-Attack, a highly transferable black-box attack framework. This framework builds on our insight that surrogate models contain a broad, high-level, cross-modally aligned semantic space. This space extends beyond final-layer outputs and provides multiple semantically consistent representations that remain underexploited by existing attacks. Within this space, O-Attack anchors aligned representations, progressively broadens semantic conditions, and optimizes perturbations through semantic consensus to promote consistent target alignment. By fully exploiting this space with the same surrogate models as M-Attack, O-Attack raises attack success rates on GPT-5.4 (29.1% $\rightarrow$ 77.2%), Claude-4.6 (42.8% $\rightarrow$ 81.6%), and Gemini-3.1 (38.2% $\rightarrow$ 80.9%). Extensive experiments across 24 MLLMs show that O-Attack outperforms six state-of-the-art methods in black-box transferability, with consistent effectiveness across prompts and improved efficiency and imperceptibility. This work exposes the practical safety risks posed by black-box adversarial attacks against frontier MLLMs, underscoring the need for more rigorous robustness evaluation and more effective defenses.

🏠 **Project page:** <https://summu77.github.io/O-Attack/> 🔗 **Code:** <https://github.com/Summu77/O-Attack>

Index Terms—Black-box adversarial attacks, adversarial transferability, cross-modal alignment, multimodal large language models.

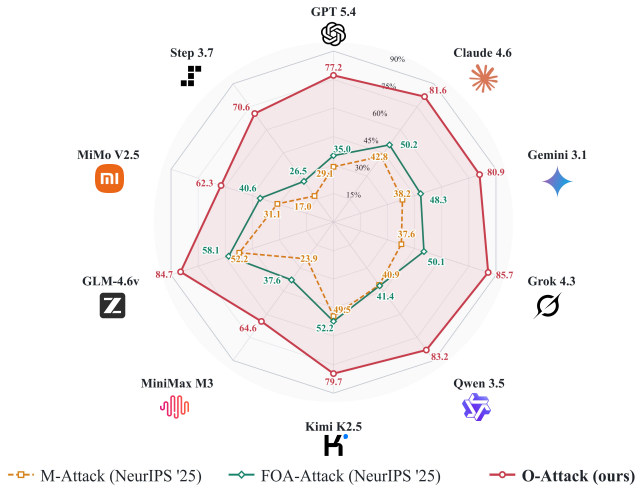


Fig. 1. Black-box attack success rates (ASR) on 10 frontier MLLMs. All methods use same surrogate models (CLIP-B/16, CLIP-B/32, and CLIP-G/14). O-Attack (ours) achieves an average ASR of 77.1%, compared with 36.2% for M-Attack [1] and 44.0% for FOA-Attack [2].

1 INTRODUCTION

MULTIMODAL large language models (MLLMs) have become foundational to modern artificial intelligence (AI) systems [3]–[5]. Their capacity to process diverse modalities jointly and reason across them enables complex multimodal understanding [6], supports domain-specific analysis [7], and underpins general-purpose agent systems [8]. This progress is driven by continued advances in proprietary models—including GPT-5.4 [9], Claude-4.6 [10], and Gemini-

3.1 [11]—alongside a rapidly maturing open-weight ecosystem exemplified by Qwen3.5 [12], GLM 4.6v [13], and Kimi K2.5 [14]. As MLLMs become increasingly embedded in everyday applications, industrial systems, and safety-critical systems, ensuring their reliability is essential for trustworthy deployment.

Among the threats to this reliability, adversarial attacks constitute a well-established paradigm: attackers add carefully crafted, human-imperceptible perturbations [15], [16] to clean inputs, thereby manipulating MLLMs into producing attacker-specified erroneous outputs or exhibiting unsafe behaviors [17]–[24]. Such attacks can be highly effective in white-box settings [25], [26], where model parameters and gradients are fully accessible, but their effectiveness often declines substantially when transferred to unseen models in black-box settings [27], [28]. This limitation is particularly consequential in today’s MLLM ecosystem, which spans heterogeneous model families, evolves through frequent version updates, and includes widely deployed proprietary frontier models accessible only through black-box interfaces. An attack optimized in a white-box setting on a small set of models may therefore generalize poorly to newly released or architecturally distinct targets, potentially leading to a substantial underestimation of real-world adversarial risk. These observations motivate a central question: *can we develop a highly transferable adversarial attack that achieves consistently high attack success rates against black-box MLLMs across diverse model families and their latest frontier versions?*

Recent black-box transfer attacks on MLLMs have taken initial steps in this direction. A common paradigm [1], [2], [29]–[33] uses widely available open-source visual encoders, such as CLIP, as surrogate models and formulates adversarial optimization as aligning an adversarial source image with a target image and target text in the surrogate representation

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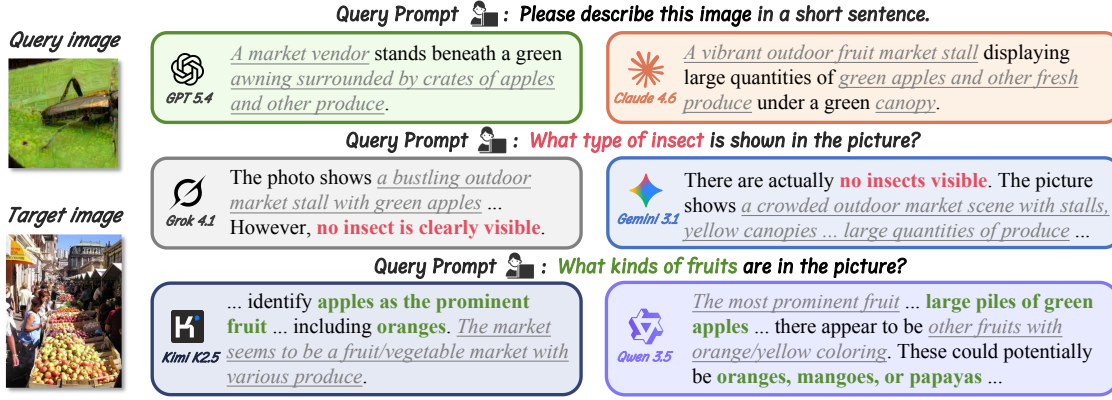


Fig. 2. An example of 0-Attack transferring across models and prompts. A single adversarial image elicits responses aligned with the target market scene from six frontier commercial MLLMs under both image-description and question-answering prompts.

space. Building on this formulation, strategies such as data augmentation and multi-surrogate ensembling encourage perturbations to capture more generalizable semantic directions rather than overfit to semantic biases specific to an individual surrogate, thereby facilitating transfer to black-box models. Despite some success, their transferability to frontier commercial MLLMs remains limited, as illustrated by M-Attack [1] and FOA-Attack [2] in Figure 1. Revisiting existing methods, we observe that they typically formulate their attack objectives using final-layer features, a natural choice for exploiting cross-modal semantics because cross-modal alignment is primarily learned at the final output stage of surrogate models. However, relying solely on this final-layer alignment may favor surrogate-specific solutions that generalize poorly to unseen MLLMs. Overall, this narrow focus on final-layer surrogate features motivates us to explore the broader cross-modal semantic space within surrogate models to improve black-box transferability.

To investigate this underexplored space, we use *Centered Kernel Alignment* (CKA) [34], [35] to compare representational geometries across layers, modalities, and models in representative CLIP surrogates and victim MLLMs. Our analysis reveals stable intra-modal geometry within each surrogate encoder and cross-modal alignment spanning a contiguous block of late visual and textual layers. Within this block, some non-final visual-textual layer pairs exhibit alignment comparable to or stronger than the final-layer pair; certain non-final surrogate-victim visual layer pairs also show higher representational similarity than their final-layer counterparts. Together, these observations indicate a **broad, high-level, cross-modally aligned semantic space that remains underexploited by existing attacks**. This space offers multiple semantically consistent representations for adversarial optimization, providing a basis for reducing reliance on the final-layer outputs of surrogate models. This finding also differs from prior efforts to improve transferability by aggregating all layers within a single modality [36], [37], as it reveals a subset of late visual and textual layers aligned in a shared high-level semantic space.

Motivated by these insights, we propose 0-Attack, a black-box attack framework that fully exploits this broad, high-level, cross-modally aligned semantic space to generate highly transferable adversarial examples. 0-Attack consists of three stages. First, *Cross-Modal Semantic Space Anchoring*

(CSA) uses CKA-based semantic trajectory analysis to select stable late visual and textual layers, capturing the semantic space identified by our analysis. Second, *Progressive Semantic Space Sampling* (PSS) explores this space through stochastic surrogate states induced by progressively widening the sampling range of dropout probabilities, moving from stable target alignment to optimization over more diverse semantic conditions. Third, *Semantic Consensus Optimization* (SCO) averages target-alignment scores over anchor layers, then maximizes their mean while penalizing their variance across surrogate models and augmented semantic views, encouraging consistent use of the semantic signals identified beyond the final layer. Together, these stages systematically exploit this semantic space to achieve strong black-box transferability. As shown in Figure 1, with both methods using exactly the same surrogate models, 0-Attack improves ASR over M-Attack on GPT-5.4 (29.1% → 77.2%), Claude-4.6 (42.8% → 81.6%), and Gemini-3.1 (38.2% → 80.9%). Figure 2 illustrates the attack’s effectiveness across both models and prompts: a single adversarial image elicits responses aligned with the target scene from six frontier commercial MLLMs under varied prompts.

Extensive experiments across 24 MLLMs validate the effectiveness of 0-Attack, demonstrating consistently stronger black-box transferability than six state-of-the-art methods. Specifically, it improves average ASR over the strongest baseline by 11.6 and 10.2 percentage points on the 10 frontier commercial MLLMs and 14 widely used MLLMs, respectively. These gains are complemented by improved efficiency and imperceptibility: 0-Attack can outperform state-of-the-art baselines with fewer surrogate models, shorter generation time, and smaller, less perceptible perturbations. We further demonstrate a practical safety risk: 0-Attack causes unsafe images to pass content moderation. Moreover, increasing model reasoning effort provides no reliable defense against the attack and can even amplify its effects. Together, these results establish 0-Attack as a highly transferable attack effective across all frontier MLLMs, motivating more rigorous robustness evaluation and the development of effective defenses. Our main contributions are summarized as follows:

- We identify a broad, high-level, cross-modally aligned semantic space spanning late visual and textual layers in surrogate models, revealing an underexploited basis for adversarial transfer beyond final-layer outputs.

- Inspired by these insights, we propose 0-Attack, which integrates semantic space anchoring, progressive sampling, and semantic consensus optimization to exploit this space for highly transferable black-box attacks.
- We demonstrate superior black-box transfer across 24 MLLMs and diverse prompts, with improved efficiency and imperceptibility. These results expose practical safety risks and motivate stronger defenses for frontier MLLMs.

2 RELATED WORK

Background on MLLMs is provided in Appendix H.

Black-Box Adversarial Attacks. Black-box adversarial attacks craft optimized, imperceptible perturbations that induce incorrect or attacker-specified predictions from a target model without access to its parameters or gradients [38], [39]. They are commonly categorized as query-based or transfer-based attacks. Query-based attacks generate adversarial examples through interactions with the target model. Representative methods either exploit confidence scores or logits, including ZOO [40], AutoZOOM [41], NES [42], Bandit Attack [43], SimBA [44], and Square Attack [45], or rely solely on hard-label predictions, including Boundary Attack [46], Sign-OPT [47], and HopSkipJumpAttack [48]. Transfer-based attacks instead optimize adversarial examples on accessible surrogate models and directly apply them to unseen target models by exploiting cross-model transferability. Early studies established this paradigm through surrogate ensembling [49], [50]. Subsequent methods improve transferability by stabilizing and diversifying the optimization trajectory, such as MI-FGSM [51], SI-NI-FGSM [52], and VMI-FGSM [53]; applying input transformations, including input diversity [54], translation invariance [55], and image admixture [56]; disrupting transferable intermediate representations through TAP [57], ILA [58], FIA [59], and NAA [60]; or modifying the surrogate’s gradient propagation using SGM [61] and LinBP [62]. Their limited reliance on target-model feedback makes transfer-based attacks a particularly practical and scalable black-box threat.

Transfer-Based Black-Box Attacks on MLLMs. Adversarial vulnerabilities have been extensively studied in models ranging from early vision systems to contemporary MLLMs [15], [16], [63]. Compared with white-box optimization [64]–[68], transfer-based black-box attacks more closely reflect real-world threat scenarios, in which attackers typically lack access to the target model’s parameters and gradients [37], [69]–[72]. Early transfer attacks [25], [29], [73], [74] optimize adversarial perturbations to align with target image or text embeddings in pretrained vision–language spaces. Recent studies have improved transferability through data augmentation, surrogate ensembling, more effective attack feature representations, and novel optimization strategies. Specifically, AttackBard [29], M-Attack [1], and X-Transfer [32] employ advanced augmentation and model-ensemble strategies, whereas M-Attack-V2 [75] further stabilizes crop-level optimization through multi-crop and auxiliary target alignment. V-Attack [31] reconsiders the choice of attack feature space by targeting value representations, while MPCAttack [33] further expands this representation space through ensembles of visual models trained under multiple learning paradigms. AnyAttack [30] learns a self-supervised

adversarial noise generator from large-scale image–text data, and FOA-Attack [2] extends this approach with patch-token-level optimal transport, whereas VCP-Attack [76] performs contrastive target alignment within dynamically constructed semantic subspaces. FRA-Attack [77] regularizes both feature alignment and surrogate gradients in the frequency domain.

However, most existing methods confine cross-modal alignment between text and vision to the final layer of the surrogate model, thereby overlooking both the model’s layer-wise semantic evolution trajectory and the broader cross-modal alignment signals available throughout its deeper layers. These limitations motivate our 0-Attack, which leverages broader cross-modal alignment signals to provide richer optimization signals for transferable attacks.

3 INSIGHTS INTO THE CROSS-MODAL ALIGNMENT

Multimodal Large Language Models (MLLMs) encode visual and textual inputs into a shared multimodal space, enabling adversarial perturbations to be optimized by aligning a source image with both target image and text representations. As the standard cross-modal alignment is exclusively performed at the output stage during model pretraining, existing attacks usually instantiate this objective using the final-layer features of surrogate models. Despite its effectiveness in white-box settings, this formulation implicitly ties the perturbation to the surrogate model’s decision boundary. As a result, the generated adversarial examples may transfer when the target model shares a similar boundary structure, but their generalization can degrade substantially across different model families. Such a last-layer-centric view ignores the vast yet under-explored semantic space within the surrogate models, limiting the optimization and hindering the discovery of more robust, cross-model adversarial perturbations.

To systematically analyze the broader, potentially exploitable cross-modal semantic space, we employ *Centered Kernel Alignment* (CKA) [34], [35] to compare the representational geometries across layers, modalities, and models. CKA compares the relational structures induced by the same ordered set of samples and permits representations with different feature dimensions. Linear CKA is invariant to orthogonal transformations and isotropic rescaling of either representation. A higher CKA score indicates greater similarity between the underlying representational geometries. Given nonzero centered feature matrices $X_i \in \mathbb{R}^{n \times d_i}$ and $X_j \in \mathbb{R}^{n \times d_j}$ whose rows correspond to the same n samples, their linear CKA similarity is defined as

$$\text{CKA}(X_i, X_j) = \frac{\|X_j^\top X_i\|_F^2}{\|X_i^\top X_i\|_F \|X_j^\top X_j\|_F} = \frac{\text{tr}(X_i^\top X_j X_j^\top X_i)}{\sqrt{\text{tr}((X_i^\top X_i)^2)} \sqrt{\text{tr}((X_j^\top X_j)^2)}}. \quad (1)$$

Building on this formulation, we randomly sample 1k image–caption pairs from COCO and extract layer-wise representations from each model. We consider two representative CLIP-based surrogate models [78], CLIP-B/16 and CLIP-G/14, together with two victim MLLMs, LLaVA-v1.6 [79] and InternVL-3.5 [80]. Specifically, for each model, we analyze representations from its final five layers with respect to three types of similarity: (i) intra-modal similarity across layers within each encoder; (ii) cross-modal similarity between the visual and text encoders; and (iii) cross-model similarity between the

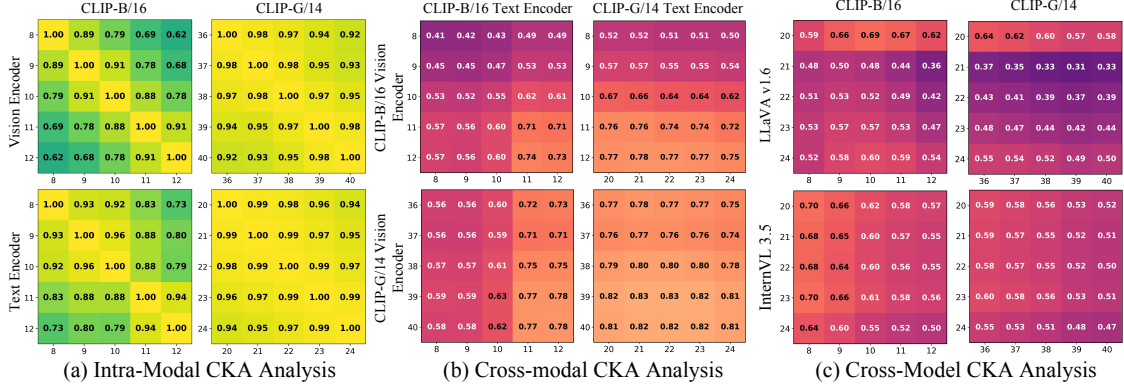


Fig. 3. Late-layer CKA analysis on surrogate CLIP models and victim MLLMs. Late-layer representations show strong intra-modal similarity, while cross-modal alignment and surrogate–victim similarity extend beyond the final layer, revealing a transferable high-level semantic space.

visual encoders of the surrogate and victim models. As shown in Figure 3, our CKA analysis yields several key findings:

- Layers nearest the final output can attain intra-modal CKA above 0.9, while similarity varies across depth and models. CLIP-G/14 exhibits a broader region of high similarity in this analysis, suggesting that high-level semantics may extend across more late layers.
- **Cross-modal alignment is not confined to the final layer but spans a contiguous block of late visual and textual layers**, where several non-final layer pairs exhibit comparable or higher CKA values (denoted by the bright orange regions) than those of the final-layer pairs.
- Surrogate–victim similarity is not maximized at the final layers. For example, the non-final late-layer pairs of CLIP-G/14 and InternVL-3.5 reach higher CKA values. This suggests that leveraging broader high-level semantics from the surrogate **may contain more transferable signals**.

Taken together, these observations suggest that, contrary to the common assumption that high-level cross-modal semantics are confined to the final layer, surrogate models preserve a stable cross-modal semantic space across their later layers. Fully exploiting this space can yield richer and more transferable optimization signals, thereby improving adversarial transferability across diverse victim models. This finding motivates 0-Attack, which explicitly leverages the high-level cross-modal semantics distributed across these layers. This design also differentiates 0-Attack from prior single-modality attacks. Although such attacks can benefit from aggregating information across early and intermediate layers [36], [37], this strategy is poorly suited to cross-modal optimization. Cross-modal alignment has not yet formed in shallow layers; incorporating their representations therefore provides little useful guidance and instead introduces noise that impedes adversarial optimization. Building on these insights, we next present the design of our 0-Attack.

4 METHODOLOGY

The insights in Section 3 motivate exploiting the broad, high-level, cross-modally aligned semantic space spanning late visual and textual layers of surrogate models. Optimizing over these layers alone, however, can still favor perturbations tied to fixed surrogate feature geometries. We therefore design 0-Attack to sample diverse semantic conditions within this space and promote consistent target alignment.

As illustrated in Figure 4, 0-Attack comprises three stages. *Stage I* uses CKA trajectories to anchor a stable region of late visual and textual layers. *Stage II* explores stochastic surrogate representations by progressively increasing the upper bound of sampled dropout probabilities. *Stage III* optimizes perturbations using an expectation–variance objective over the resulting semantic conditions. We also adopt data augmentation and surrogate ensembling as standard strategies for improving attack transferability, following prior work [1], [2], [29], [30], [33], [75].

4.1 Preliminaries

We first formalize the targeted black-box attack setting. Given a source image $x_s \in [0, 1]^d$, a target image $x_t \in [0, 1]^d$, and an ℓ_∞ budget ϵ , we seek a perturbation δ with $\|\delta\|_\infty \leq \epsilon$ that redirects unknown victim MLLMs toward the target image’s semantics. The returned adversarial image is $x_{\text{adv}} = \Pi_{[0,1]^d}(x_s + \delta)$, where Π denotes elementwise clipping; since $x_s \in [0, 1]^d$, this also ensures $\|x_{\text{adv}} - x_s\|_\infty \leq \epsilon$. We construct source and target caption banks, $\mathcal{C}_s = \{c_s^i\}_{i=1}^N$ and $\mathcal{C}_t = \{c_t^i\}_{i=1}^N$, containing N GPT-generated descriptions of each image. We optimize only on accessible surrogate models, indexed by $\mathcal{M} = \{1, \dots, M\}$, without querying victim models during optimization [1], [2], [29], [30]. Each surrogate m consists of a visual encoder f_m and a text encoder g_m with pretrained projections into a shared embedding space. All surrogate parameters remain frozen. Image inputs are expressed on the $[0, 1]$ scale; encoder preprocessing includes resizing, pixel clipping, and normalization.

4.2 Stage I: Cross-Modal Semantic Space Anchoring

The cross-modal alignment observed in Section 3 motivates selecting late layers with stable representational geometry. For each surrogate m , let $h_{m,\ell}^v(x) \in \mathbb{R}^{d_m^v}$ be the visual CLS token after transformer block ℓ , and let $h_{m,\ell}^t(c) \in \mathbb{R}^{d_m^t}$ be the corresponding textual end-of-sequence (EOS) token. We use these unprojected representations for offline CKA analysis with dropout disabled. For each modality $a \in \{v, t\}$, stack the token representations of n_a calibration inputs as rows of $H_{m,\ell}^a \in \mathbb{R}^{n_a \times d_m^a}$. We compare each layer with the final block L_m^a using the linear CKA from Section 3:

$$q_{m,\ell}^a = \text{CKA}\left(\bar{H}_{m,\ell}^a, \bar{H}_{m,L_m^a}^a\right), \quad \bar{H}_{m,\ell}^a = \left(I_{n_a} - \frac{1}{n_a} \mathbf{1}_{n_a} \mathbf{1}_{n_a}^\top\right) H_{m,\ell}^a. \quad (2)$$

Here, I_{n_a} is the identity matrix and $\mathbf{1}_{n_a}$ is the all-ones vector. CKA is evaluated on nonzero centered representations. The

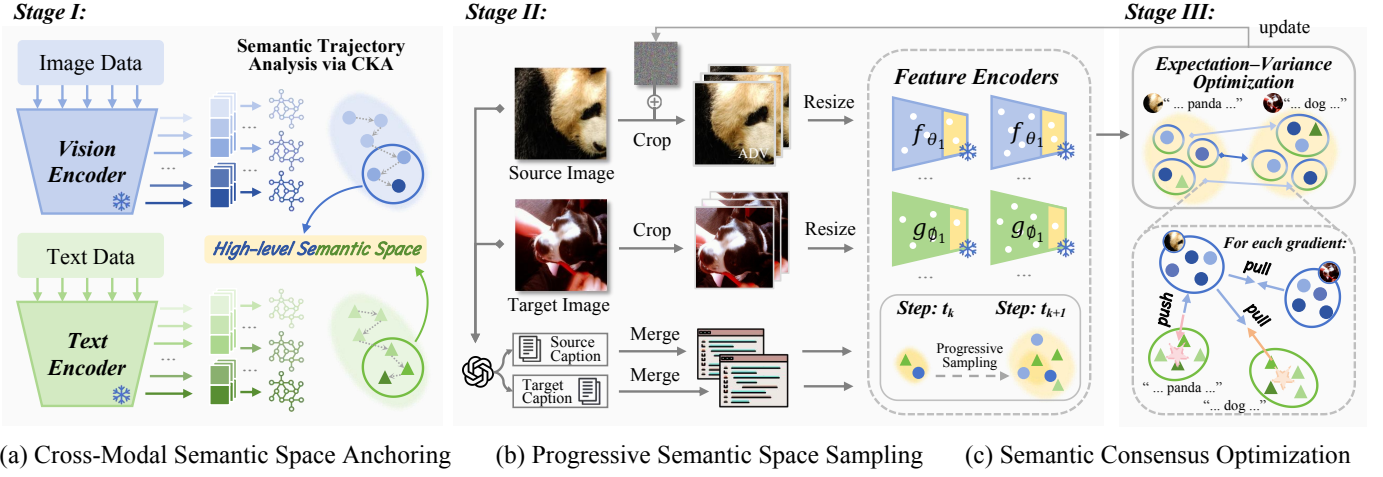


Fig. 4. Overview of 0-Attack. (a) CKA trajectories guide the selection of late visual and textual anchor layers. (b) Progressive dropout sampling explores diverse semantic conditions within the anchored space. (c) Semantic consensus optimization promotes high mean target alignment and low variance across surrogates and augmented views. Here, f_{θ_1} and g_{ϕ_1} denote a surrogate's frozen visual and text encoders.

trajectories guide the offline choice of a contiguous late-layer anchor set $S_m^a = \{\ell_m^a, \dots, L_m^a\}$. Similarity thresholds used to identify this stable region can vary across encoders and surrogate models; the selected ranges remain fixed during optimization. This within-encoder diagnostic complements the cross-modal analysis in Section 3; it does not itself establish alignment between modalities.

To place the selected representations in the surrogate's shared embedding space, we apply the pretrained final LayerNorm and projection of the corresponding encoder:

$$z_{m,\ell}^a(u) = \text{Norm}(P_m^a \text{LN}_m^a(h_{m,\ell}^a(u))), \quad a \in \{v, t\}. \quad (3)$$

Here, u is an image or caption, and $\text{Norm}(w) = w/\|w\|_2$ for nonzero w . We assume the vectors normalized below are nonzero. Within each encoder, the same final LayerNorm and projection are applied to every anchor layer. Both modalities are projected into $\mathbb{R}^{d_m}$, so their inner products are well defined even when $d_m^v \neq d_m^t$. The resulting feature sets $\mathcal{Z}_m^a(u) = \{z_{m,\ell}^a(u) : \ell \in S_m^a\}$ provide the representations used in the next two stages.

4.3 Stage II: Progressive Semantic Space Sampling

Optimizing only the deterministic anchor features may overfit the perturbation to a limited set of surrogate representations. PSS introduces stochastic variation by enabling dropout on attention weights and feed-forward outputs throughout both encoders [81]–[83], while retaining the anchor layers selected in Stage I. Let ξ collect the dropout masks for one encoded input. We denote its token and projected features by $h_{m,\ell,\xi}^a$ and $z_{m,\ell,\xi}^a$, respectively, with the projection and normalization defined in Eq. (3).

For T iterations and $t \in \{0, \dots, T-1\}$, the probability cap follows the linear schedule

$$\bar{p}_{m,t} = p_m^{\text{start}} + \frac{t}{T-1}(p_m^{\text{end}} - p_m^{\text{start}}). \quad (4)$$

The initial and final caps lie in $[0, 1)$ and may differ across surrogates. Progressive sampling uses $p_m^{\text{start}} \leq p_m^{\text{end}}$, reversing these endpoints defines the decreasing schedule examined in the ablation study. Both encoders share this cap. At each

iteration, we independently sample a probability for each surrogate, modality, and block type:

$$p_{m,t}^{a,b} \sim \mathcal{U}(0, \bar{p}_{m,t}), \quad a \in \{v, t\}, \quad b \in \{\text{attn}, \text{ffn}\}. \quad (5)$$

When $\bar{p}_{m,t} = 0$, the sampled probabilities are zero. Each probability is held fixed across the corresponding block type in all layers and views of its encoder for that iteration. Dropout masks are sampled independently for different inputs and forward passes, conditional on these probabilities. Under the progressive schedule, the cap increases monotonically, while individual sampled probabilities need not. This schedule limits early stochastic variation, then broadens the range of dropout intensities while retaining small probabilities.

At each iteration, we construct K semantic views using image augmentation and caption mixing [1], [2], [29], [30]. For view k , independently sampled crop-resize transforms A_k, B_k produce $\tilde{x}_{\text{adv}}^k = A_k(x_s + \delta)$ and $\tilde{x}_t^k = B_k(x_t)$; pixel clipping is applied subsequently during encoder preprocessing. The same augmented inputs are evaluated by all surrogates. We also draw two captions from each bank and mix their words: a random fraction $\beta \sim \mathcal{U}(0, 1)$ determines a prefix of the first caption and a complementary suffix of the second, with both word counts rounded down. Source and target mixtures $\tilde{c}_s^k, \tilde{c}_t^k$ are sampled independently. For each model and view, $\xi_{\text{adv}}, \xi_{\text{img}}, \xi_{\text{src}}, \xi_{\text{tgt}}$ denote the separate dropout realizations for the two images and two captions; their model, view, and iteration indices are suppressed for readability. All anchor features of an input are obtained from its same stochastic forward pass.

4.4 Stage III: Semantic Consensus Optimization

SCO promotes high target alignment and low dispersion across the sampled model-view pairs. All quantities below refer to the current iteration, whose index is suppressed except in the loss. We first average and renormalize each surrogate's normalized textual anchor features:

$$z_{m,\xi}^t(c) = \text{Norm}\left(\frac{1}{|S_m^t|} \sum_{\ell \in S_m^t} z_{m,\ell,\xi}^t(c)\right). \quad (6)$$

TABLE 1
Black-box adversarial attack performance on 10 frontier commercial MLLMs and 14 widely used MLLMs.

Model	AnyAttack		COA		M-Attack		FOA-Attack		M-Attack-V2		MPCAttack		O-Attack (ours)	
	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim
<i>10 frontier commercial MLLMs</i>														
GPT 5.4	2.3	0.071	1.8	0.026	29.1	0.325	35.0	0.381	46.5	0.485	43.5	0.439	77.2	0.650
Claude 4.6	1.4	0.044	0.9	0.014	42.8	0.452	50.2	0.495	76.4	0.643	76.7	0.627	81.6	0.693
Gemini 3.1	1.7	0.055	0.0	0.028	38.2	0.419	48.3	0.482	66.7	0.581	77.6	0.649	80.9	0.678
Grok 4.3	1.5	0.054	0.9	0.015	37.6	0.469	50.1	0.512	79.6	0.637	66.3	0.620	85.7	0.700
Qwen 3.5	1.7	0.065	0.7	0.030	40.9	0.439	41.4	0.466	77.9	0.664	77.7	0.663	83.2	0.668
Kimi K2.5	3.9	0.064	0.0	0.024	49.5	0.504	52.2	0.548	67.8	0.628	76.2	0.632	79.7	0.652
MiniMax M3	2.6	0.078	0.5	0.031	23.9	0.314	37.6	0.420	51.4	0.516	45.5	0.452	64.6	0.580
GLM-4.6V	1.3	0.058	0.6	0.017	52.2	0.529	58.1	0.553	83.0	0.679	81.6	0.672	84.7	0.688
MiMo V2.5	2.9	0.082	0.4	0.026	31.1	0.358	40.6	0.404	59.8	0.566	53.6	0.495	62.3	0.571
Step 3.7	1.2	0.053	0.3	0.037	17.0	0.256	26.5	0.316	46.3	0.430	47.2	0.439	70.6	0.607
Avg. (10)	2.1	0.062	0.6	0.025	36.2	0.407	44.0	0.458	<u>65.5</u>	<u>0.583</u>	64.6	0.569	77.1	0.648
<i>14 widely used MLLMs</i>														
Llama 4	3.3	0.071	0.8	0.056	55.8	0.513	55.1	0.550	79.0	0.664	76.7	0.642	80.5	0.688
NEX N2	2.5	0.083	1.3	0.042	58.6	0.536	67.1	0.577	80.4	0.686	81.6	0.704	92.0	0.722
LLaVA v1.6	3.3	0.087	0.6	0.039	73.9	0.640	77.6	0.672	89.5	0.774	91.3	0.775	95.5	0.796
MiniCPM V4.5	3.1	0.097	0.7	0.028	63.3	0.563	63.0	0.577	79.1	0.691	80.3	0.674	90.7	0.734
Gemma 3	0.3	0.100	1.1	0.040	26.6	0.330	36.7	0.380	48.2	0.495	56.0	0.522	71.4	0.628
InternVL 3.5	4.1	0.088	0.0	0.033	61.0	0.541	64.3	0.590	84.1	0.711	85.1	0.704	90.4	0.737
DeepSeekVL 2	2.6	0.078	0.9	0.033	62.9	0.599	76.9	0.647	91.0	0.749	92.6	0.757	95.1	0.763
Molmo 2	0.7	0.063	0.2	0.018	42.5	0.434	41.2	0.440	61.5	0.591	63.8	0.611	77.5	0.650
Llama-Vision	3.9	0.096	0.1	0.035	57.5	0.563	64.5	0.589	77.0	0.701	84.7	0.728	88.4	0.758
Mistral 3.5	1.7	0.069	0.1	0.033	34.7	0.382	35.3	0.388	48.7	0.476	38.8	0.401	66.2	0.601
Qwen3 VL	4.7	0.072	0.7	0.031	57.8	0.544	66.2	0.567	86.1	0.694	80.0	0.681	88.3	0.730
Seed 2.0	1.7	0.063	0.0	0.020	12.6	0.229	16.3	0.269	15.7	0.269	23.4	0.297	49.2	0.510
Nova 2	2.2	0.075	0.6	0.030	40.4	0.457	44.0	0.511	68.7	0.630	71.0	0.635	73.1	0.645
Nemotron 2	3.1	0.084	0.3	0.022	40.0	0.459	47.8	0.482	65.0	0.612	75.3	0.667	84.9	0.687
Avg. (14)	2.7	0.080	0.5	0.033	49.1	0.485	54.0	0.517	69.6	0.625	<u>71.5</u>	<u>0.628</u>	81.7	0.689

For $\ell \in \mathcal{S}_m^v$ and $k \in \{1, \dots, K\}$, the visual alignment and text-guided contrast scores are

$$\begin{aligned}
 r_{m,\ell,k}^v(\delta) &= \left\langle z_{m,\ell,\xi_{\text{adv}}}^v(\tilde{x}_{\text{adv}}^k), z_{m,\ell,\xi_{\text{img}}}^v(\tilde{x}_t^k) \right\rangle, \\
 r_{m,\ell,k}^t(\delta) &= \left\langle z_{m,\ell,\xi_{\text{adv}}}^v(\tilde{x}_{\text{adv}}^k), \tilde{z}_{m,\ell,\xi_{\text{tgt}}}^t(\tilde{c}_t^k) \right\rangle \\
 &\quad - \left\langle z_{m,\ell,\xi_{\text{adv}}}^v(\tilde{x}_{\text{adv}}^k), \tilde{z}_{m,\ell,\xi_{\text{src}}}^t(\tilde{c}_s^k) \right\rangle.
 \end{aligned} \quad (7)$$

The visual score attracts the perturbed image toward the target image, while the text score favors the target caption over the source caption. For each score type $a \in \{v, t\}$, we first average scores over visual anchor layers:

$$\bar{r}_{m,k}^a(\delta) = \frac{1}{|\mathcal{S}_m^v|} \sum_{\ell \in \mathcal{S}_m^v} r_{m,\ell,k}^a(\delta). \quad (8)$$

The empirical moments across MK model-view pairs are

$$\begin{aligned}
 \hat{\mu}_a(\delta) &= \frac{1}{MK} \sum_{m=1}^M \sum_{k=1}^K \bar{r}_{m,k}^a(\delta), \\
 \hat{\sigma}_a^2(\delta) &= \frac{1}{MK} \sum_{m=1}^M \sum_{k=1}^K (\bar{r}_{m,k}^a(\delta) - \hat{\mu}_a(\delta))^2.
 \end{aligned} \quad (9)$$

Each surrogate receives equal weight regardless of its number of anchor layers. The variance measures dispersion of the layer-averaged scores; it does not directly penalize

differences between individual layers. Shared views and per-iteration dropout probabilities also mean that the MK scores need not be independent. We minimize the following sampled loss subject to $\|\delta\|_\infty \leq \epsilon$:

$$\begin{aligned}
 \mathcal{L}_t(\delta) &= -\hat{\mu}_v(\delta) + \lambda_v \hat{\sigma}_v^2(\delta) \\
 &\quad + \alpha [-\hat{\mu}_t(\delta) + \lambda_t \hat{\sigma}_t^2(\delta)],
 \end{aligned} \quad (10)$$

where $\alpha \geq 0$ weights the text-guided term and $\lambda_v, \lambda_t \geq 0$ control the variance penalties. The mean terms promote target alignment, while the variance terms penalize disagreement across sampled conditions. Appendix A establishes an exact consensus interpretation, a local sensitivity interpretation under smoothness assumptions, and a conditional bound under distribution shift. Starting from $\delta_0 = 0$, we take an Adam step on $\mathcal{L}_t$ and project the perturbation onto $[-\epsilon, \epsilon]^d$ at each iteration (see Algorithm 1 in the appendix).

5 EXPERIMENTS

5.1 Settings

Datasets and Models. Following prior works [1], [2], [29], [33], [75], we use 1,000 clean images from the NIPS 2017 Adversarial Attacks and Defenses Competition dataset as sources and randomly sample 1,000 MSCOCO validation images [84] as targets. Each source and target image is paired with five

TABLE 2
Performance under different perturbation budgets ϵ , with normalized ℓ_1 and ℓ_2 perturbation magnitudes.

ϵ	Method	GPT 5.4		Claude 4.6		Gemini 3.1		Grok 4.3		Qwen 3.5		Kimi K2.5		Average		Perturbation	
		ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	$\ell_1^{\text{norm}} \downarrow$	$\ell_2^{\text{norm}} \downarrow$
8/255	M-Attack	16.0	0.212	4.0	0.143	21.0	0.249	19.0	0.278	11.0	0.214	2.0	0.079	12.2	0.196	0.0207	0.0233
	FOA-Attack	14.0	0.227	6.0	0.131	21.0	0.272	26.0	0.325	16.0	0.231	5.0	0.096	14.7	0.214	<u>0.0208</u>	<u>0.0234</u>
	M-Attack-V2	<u>32.0</u>	<u>0.337</u>	<u>23.0</u>	<u>0.320</u>	27.0	0.371	48.0	0.506	49.0	0.503	20.0	0.264	33.2	<u>0.383</u>	0.0233	0.0254
	MPCAttack	25.0	0.309	10.0	0.212	48.0	0.469	54.0	0.509	46.0	0.492	23.0	0.266	34.3	0.376	0.0235	0.0257
	O-Attack (ours)	52.0	0.494	40.0	0.436	53.0	0.509	57.0	0.557	59.0	0.569	25.0	0.342	47.67	0.484	0.0224	0.0247
12/255	M-Attack	28.0	0.298	38.0	0.364	30.0	0.354	35.0	0.389	29.0	0.338	18.0	0.215	29.7	0.326	0.0282	0.0324
	FOA-Attack	33.0	0.345	39.0	0.401	36.0	0.387	46.0	0.462	37.0	0.380	15.0	0.209	34.3	0.364	<u>0.0285</u>	<u>0.0326</u>
	M-Attack-V2	<u>41.0</u>	<u>0.423</u>	50.0	<u>0.521</u>	54.0	0.513	68.0	0.617	69.0	0.623	52.0	0.517	55.7	0.536	0.0320	0.0356
	MPCAttack	35.0	0.401	<u>56.0</u>	0.511	70.0	0.614	<u>69.0</u>	<u>0.623</u>	75.0	0.619	<u>56.0</u>	<u>0.528</u>	<u>60.2</u>	<u>0.549</u>	0.0329	0.0366
	O-Attack (ours)	64.0	0.577	63.0	0.592	<u>67.0</u>	<u>0.612</u>	72.0	0.626	<u>71.0</u>	0.636	67.0	0.582	67.33	0.604	0.0306	0.0345
16/255	M-Attack	29.1	0.325	42.8	0.452	38.2	0.419	37.6	0.469	40.9	0.439	49.5	0.504	39.7	0.435	0.0340	0.0398
	FOA-Attack	35.0	0.381	50.2	0.495	48.3	0.482	50.1	0.512	41.4	0.466	52.2	0.548	46.2	0.481	<u>0.0345</u>	<u>0.0403</u>
	M-Attack-V2	<u>46.5</u>	<u>0.485</u>	76.4	<u>0.643</u>	66.7	0.581	<u>79.6</u>	<u>0.637</u>	<u>77.9</u>	<u>0.664</u>	67.8	0.628	69.2	<u>0.606</u>	0.0393	0.0446
	MPCAttack	43.5	0.439	<u>76.7</u>	0.627	<u>77.6</u>	<u>0.649</u>	66.3	0.620	77.7	0.663	<u>76.2</u>	<u>0.632</u>	<u>69.7</u>	0.605	0.0417	0.0469
	O-Attack (ours)	77.2	0.650	81.6	0.693	80.9	0.678	85.7	0.700	83.2	0.668	79.7	0.652	81.4	0.673	0.0370	0.0427

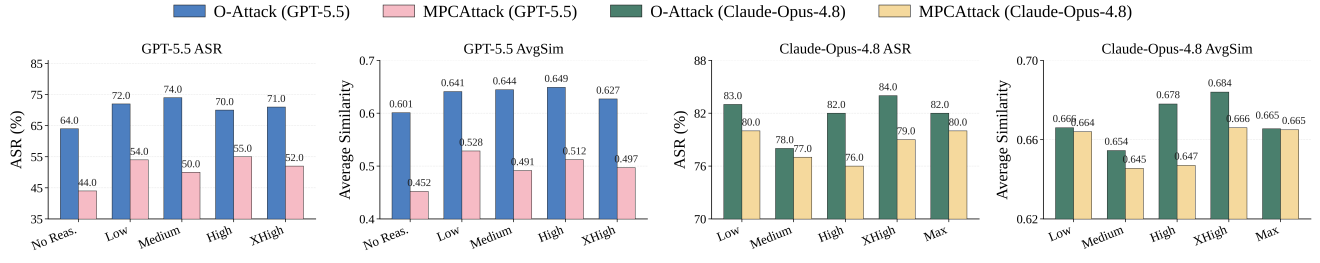


Fig. 5. Attack success rate (ASR) and average similarity (AvgSim) under different levels of reasoning effort. The results compare O-Attack and MPCAttack on GPT-5.5 and Claude Opus 4.8, with reasoning effort ranging from no reasoning to extra-high reasoning.

TABLE 3

Comparison of transfer-based attacks across optimization modalities, feature layers, sampling strategies, and surrogate models. I: image; T+I: text and image; Fix./Prog.: fixed/progressive sampling.

Method	Publication	Mod.	Layer(s)	Samp.	Surrogate Model
AnyAttack	CVPR'25	I	Last One	Fix.	CLIP-B/32; EVA-02-L/14; ViT-B/16
COA	CVPR'25	T+I	Last One	Fix.	CLIP-B/32
M-Attack	NeurIPS'25	I	Last One	Fix.	CLIP-B/32; CLIP-B/16; CLIP-G/14
FOA-Attack	NeurIPS'25	I	Last One	Fix.	CLIP-B/32; CLIP-B/16; CLIP-G/14
M-Attack-V2	Arxiv'26	I	Last One	Fix.	CLIP-B/32; CLIP-B/16; CLIP-G/14; CLIP-B/32 (LAION)
MPCAttack	CVPR'26	T+I	Last One	Fix.	CLIP-B/32; CLIP-B/16; CLIP-G/14; DINOv2; InternVL3-1B
O-Attack	Ours	T+I	Late layers	Prog.	CLIP-B/32; CLIP-B/16; CLIP-G/14

diverse captions generated by a large language model. To assess whether each adversarial image remains effective across prompts, we sample 300 source and 300 target images from VQA v2.0 [85], together with a subset of questions associated with the source images. We further use UnsafeBench [86] to expose practical safety risks in image moderation. In total, we evaluate 10 frontier commercial MLLMs and 14 widely used MLLMs; the complete model list is provided in Appendix D.

Implementation Settings. We compare O-Attack with six transfer-based attacks: AnyAttack [30], COA [74], M-Attack [1], FOA-Attack [2], M-Attack-V2 [75], and MPCAttack [33]. Table 3 summarizes their key differences. Surrogate model identifiers are listed in Appendix Table 10. We use the original implementations. Unless otherwise stated, all

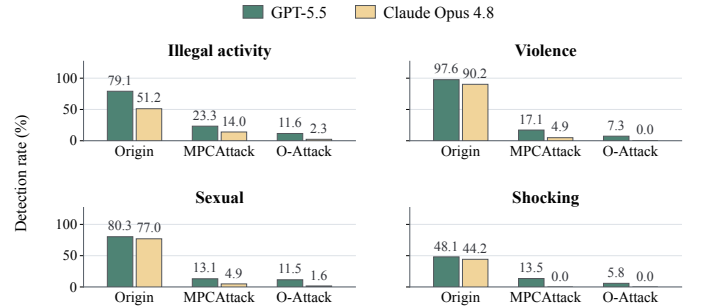


Fig. 6. UnsafeBench detection rates across four unsafe-content categories. Lower values indicate stronger moderation evasion.

methods use an ℓ_∞ perturbation budget $\epsilon = 16/255$ and $T = 300$ optimization steps. O-Attack uses the same three surrogate models as M-Attack and FOA-Attack: CLIP ViT-B/16, ViT-B/32, and ViT-g-14-laion2B-s12B-b42K. The fixed anchor ranges use the final 15 visual and 5 textual blocks of CLIP-G/14, and the final 4 visual and 2 textual blocks of both CLIP-B/16 and CLIP-B/32. The linear dropout cap schedule uses $(p_m^{\text{start}}, p_m^{\text{end}}) = (0.10, 0.15)$ for CLIP-G/14 and $(0.05, 0.10)$ for CLIP-B/16 and CLIP-B/32.

Evaluation Metrics. Following [1], [2], [33], [75], we adopt an LLM-as-a-judge protocol [87] to compute GPTScore [88]: the victim MLLM generates captions for the adversarial and target images, and GPTScore measures their semantic similarity using a fixed evaluation prompt. Attack success



Fig. 7. Qualitative safety-moderation failures on UnsafeBench. Adversarial images generated by 0-Attack cause GPT-5.5 and Claude Opus 4.8 to classify unsafe content from four representative categories—illegal activity, shocking, sexual, and violence—as safe.

TABLE 4
VQA v2 evaluation under different attacks on six frontier commercial MLLMs. Lower Acc* and EM indicate stronger attacks.

Method	GPT 5.4		Claude 4.6		Gemini 3.1		Grok 4.1		Qwen 3.5		Kimi K2.5		Avg. (6)	
	Acc* (↓)	EM (↓)	Acc* (↓)	EM (↓)	Acc* (↓)	EM (↓)	Acc* (↓)	EM (↓)	Acc* (↓)	EM (↓)	Acc* (↓)	EM (↓)	Acc* (↓)	EM (↓)
Clean	69.6	55.1	45.2	38.8	67.6	55.1	45.2	35.7	78.8	68.4	57.5	49.0	60.7	50.3
M-Attack	35.9	25.5	16.1	14.3	42.3	34.4	22.4	17.3	36.7	29.6	22.2	17.7	29.3	23.1
FOA-Attack	36.8	25.9	14.2	11.9	42.4	34.4	24.1	18.0	35.9	27.6	23.6	18.4	29.5	22.7
M-Attack-V2	34.2	25.9	10.8	8.5	34.4	27.6	21.4	15.3	31.0	22.8	13.1	9.6	24.1	18.3
MPCAttack	31.7	23.8	10.9	8.2	28.7	22.8	19.2	13.6	26.4	21.1	15.0	12.2	22.0	17.0
O-Attack	21.9	16.3	7.4	6.5	22.3	15.3	14.1	10.2	20.5	15.6	10.3	8.5	16.1	12.1

rate (ASR) is the percentage of samples with GPTScore above 0.5 [2], [33], while average similarity (AvgSim) is the mean GPTScore across all samples. The prompts and results under other thresholds are provided in Appendix E and Appendix C. For VQA, we report soft accuracy (Acc*) and Exact Match (EM), where lower values indicate stronger attacks. On UnsafeBench, attack success denotes a ground-truth unsafe image being incorrectly classified as safe.

5.2 Main Results

Results on frontier commercial MLLMs. Table 1 reports black-box transfer performance on 10 frontier commercial MLLMs. 0-Attack achieves the best result on every victim model, with an average ASR of 77.1% and AvgSim of 0.648. The strongest baseline reaches 65.5% ASR and 0.583 AvgSim, leaving gains of 11.6 percentage points and 0.065, respectively. These results show that the transferable semantic signals exploited by 0-Attack remain effective across rapidly evolving proprietary and native multimodal model families. **Results on widely used MLLMs.** The second part of Table 1 extends the evaluation to 14 widely used MLLMs. 0-Attack again ranks first on all models, improving the average ASR from 71.5% to 81.7% and AvgSim from 0.628 to 0.689 over the strongest baseline. Consistent improvements across all 24 evaluated MLLMs indicate that the attack does not depend on a narrow surrogate-victim architectural match.

Results under constrained perturbation budgets. Table 2 evaluates $\epsilon \in \{8, 12, 16\}/255$ on six frontier commercial MLLMs. At matched budgets, 0-Attack achieves average ASRs of 47.7%, 67.3%, and 81.4%, exceeding the strongest

baselines (34.3%, 60.2%, and 69.7%, respectively). It also achieves comparable attack success with smaller perturbations: at 12/255, its 67.3% ASR approaches the strongest baseline’s 69.7% at 16/255, with a 25% lower budget and smaller normalized ℓ_1 and ℓ_2 magnitudes. Meanwhile, AvgSim rises with the budget, indicating that additional perturbation capacity translates into stronger target alignment.

5.3 Exploratory Studies

Generalization across prompts. Table 4 evaluates whether adversarial examples optimized for semantic transfer remain effective under question answering prompts. On VQA v2.0, 0-Attack reduces the six-model average soft accuracy from 60.7% on clean images to 16.1%, and reduces EM from 50.3% to 12.1%. It also outperforms the strongest attack baseline, MPCAttack, by 5.9 percentage points in Acc* and 4.9 points in EM. The consistent reduction across all six victims shows that the attack transfers beyond free-form captioning to prompts that require task-specific visual reasoning.

Effect of reasoning effort. Figure 5 compares 0-Attack and MPCAttack across five reasoning-effort levels on GPT-5.5 and Claude Opus 4.8. Additional reasoning does not consistently reduce ASR or AvgSim. For 0-Attack on GPT-5.5, high-effort reasoning raises ASR from 64% to 70% and AvgSim from 0.601 to 0.649 compared with no reasoning. Figure 9 illustrates this risk: an attack that fails without reasoning succeeds under high effort, with the same image eliciting a target-aligned description. These results show that increasing reasoning effort alone offers no reliable defense and can even amplify the induced semantic errors.

TABLE 5

Component ablation of 0-Attack on six frontier commercial MLLMs. CSA: cross-modal semantic space anchoring (Space: visual encoders; Text: textual encoders). PSS: progressive semantic space sampling; SCO: semantic consensus optimization.

Method	CSA		PSS	SCO	GPT 5.4		Claude 4.6		Gemini 3.1		Grok 4.3		Qwen 3.5		Kimi K2.5		Avg.(6)	
	Space	Text			ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim
O-Attack	✓	✓	✓	✓	77.2	0.650	81.6	0.693	80.9	0.678	85.7	0.700	83.2	0.668	79.7	0.652	81.4	0.673
O-Attack	✓	✓	×	✓	59.0	0.541	77.0	0.675	66.0	0.622	72.0	0.632	76.0	0.661	77.0	0.650	71.2	0.630
O-Attack	✓	×	✓	✓	52.0	0.520	70.0	0.625	68.0	0.593	59.0	0.579	74.0	0.617	62.0	0.529	64.2	0.577
O-Attack	×	✓	×	✓	41.0	0.451	68.0	0.585	52.0	0.532	65.0	0.589	76.0	0.624	62.0	0.553	60.7	0.556
O-Attack	×	×	×	✓	39.0	0.442	67.0	0.584	53.0	0.534	65.0	0.574	62.0	0.583	73.0	0.625	59.8	0.557
M-Attack	×	×	×	×	29.1	0.325	42.8	0.452	38.2	0.419	37.6	0.469	40.9	0.439	49.5	0.504	39.7	0.435

TABLE 6

Ablation study of surrogate CLIP backbones on six frontier commercial MLLMs. B/32(l) denotes the LAION version of CLIP-B/32.

Method	Surrogate CLIP Models				GPT 5.4		Claude 4.6		Gemini 3.1		Grok 4.3		Qwen 3.5		Kimi K2.5		Avg.(6)	
	G/14	B/32	B/16	B/32(l)	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim
M-Attack-V2	✓	✓	✓	✓	46.5	0.485	76.4	0.643	66.7	0.581	79.6	0.637	77.9	0.664	67.8	0.628	69.2	0.606
FOA-Attack	✓	✓	✓	×	35.0	0.381	50.2	0.495	48.3	0.482	50.1	0.512	41.4	0.466	52.2	0.548	46.2	0.481
O-Attack	✓	✓	✓	✓	82.0	0.662	83.0	0.687	78.0	0.660	86.0	0.713	84.0	0.693	80.0	0.655	82.2	0.678
O-Attack	✓	✓	×	×	77.2	0.650	81.6	0.693	80.9	0.678	85.7	0.700	83.2	0.668	79.7	0.652	81.4	0.673
O-Attack	✓	✓	×	×	67.6	0.587	82.7	0.676	68.5	0.589	78.4	0.640	74.6	0.629	77.4	0.634	74.9	0.626
O-Attack	✓	×	✓	×	67.6	0.571	66.9	0.584	74.8	0.633	76.6	0.618	67.4	0.549	68.5	0.560	70.3	0.586
O-Attack	✓	×	×	×	48.6	0.434	56.5	0.500	64.0	0.546	71.3	0.580	61.0	0.510	58.5	0.512	60.0	0.514

TABLE 7

Ablation study of source augmentation crop number K on six frontier commercial MLLMs. #Num. denotes the number of surrogate models used by each method, and Time reports the end-to-end testing time per image on a single RTX 4090 GPU.

Method	Settings	#Num.	Time	GPT 5.4		Claude 4.6		Gemini 3.1		Grok 4.3		Qwen 3.5		Kimi K2.5		Avg. (6)	
				ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim
MPCAttack	—	6	~79s	43.5	0.439	76.7	0.627	77.6	0.649	66.3	0.620	77.7	0.663	76.2	0.632	69.7	0.605
O-Attack	$K = 3$	3	~86s	62.0	0.574	74.0	0.640	74.0	0.643	79.0	0.666	74.0	0.648	67.0	0.611	71.7	0.630
O-Attack	$K = 5$	3	~145s	64.0	0.594	71.0	0.630	79.0	0.661	82.0	0.692	80.0	0.672	75.0	0.639	75.2	0.648
O-Attack	$K = 8$	3	~210s	69.0	0.607	79.0	0.667	76.0	0.653	84.0	0.701	78.0	0.665	78.0	0.658	77.3	0.658
O-Attack	$K = 10$	3	~245s	77.2	0.650	81.6	0.693	80.9	0.678	85.7	0.700	83.2	0.668	79.7	0.652	81.4	0.673
M-Attack-V2	$K = 10$	4	~283s	46.5	0.485	76.4	0.643	66.7	0.581	79.6	0.637	77.9	0.664	67.8	0.628	69.2	0.606
FOA-Attack	$C = [3, 5, 8, 10]$	3	~284s	35.0	0.381	50.2	0.495	48.3	0.482	50.1	0.512	41.4	0.466	52.2	0.548	46.2	0.481

TABLE 8

Ablation study of progressive dropout sampling in 0-Attack.

Dropout cap		GPT 5.4		Claude 4.6		Gemini 3.1		Grok 4.3		Qwen 3.5		Kimi K2.5		Avg. (6)	
G/14	B/16, B/32	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim	ASR	AvgSim
0.05 → 0.20	0.0 → 0.15	66.0	0.613	76.0	0.660	78.0	0.671	76.0	0.672	74.0	0.654	66.0	0.612	72.7	0.647
0.05 → 0.10	0.0 → 0.05	68.0	0.584	77.0	0.681	75.0	0.667	79.0	0.679	75.0	0.653	70.0	0.625	74.0	0.648
0.15 → 0.10	0.10 → 0.05	73.0	0.619	76.0	0.648	74.0	0.640	77.0	0.660	76.0	0.663	79.0	0.657	75.8	0.648
0.10 → 0.15	0.05 → 0.10	77.2	0.650	81.6	0.693	80.9	0.678	85.7	0.700	83.2	0.668	79.7	0.652	81.4	0.673

Transfer to safety-critical image moderation. We next study whether 0-Attack extends to a practically harmful setting. On UnsafeBench [86], the victim model must decide whether an image contains a specified unsafe category, and an attack succeeds when unsafe content is incorrectly judged safe. Figure 6 reports the detection rates for illegal activity, violence, sexual, and shocking content, where lower values indicate stronger moderation evasion. Averaged over the four categories, 0-Attack reduces the detection rate from 76.3% to 9.1% on GPT-5.5 and from 65.7% to 1.0% on Claude Opus 4.8. Figure 7 illustrates these failures across all four categories: the models misclassify adversarial images as safe and support their decisions with benign scene descriptions. For example, violent content is described as people walking near

a wall or a decorated kitchen. Together, the quantitative and qualitative results show that the transferable perturbation can suppress safety-relevant visual semantics and bypass practical image-moderation decisions.

Semantic structure of adversarial perturbations. Figure 8 provides a case analysis of three source–target pairs by visualizing both the adversarial examples and their amplified perturbation textures. The perturbations produced by AnyAttack, M-Attack, and FOA-Attack are comparatively fragmented or dominated by irregular high-frequency patterns. In contrast, 0-Attack forms cleaner target-related contours and more coherent object-level structures, while the adversarial images continue to preserve the overall appearance of the source. This suggests that 0-Attack concentrates

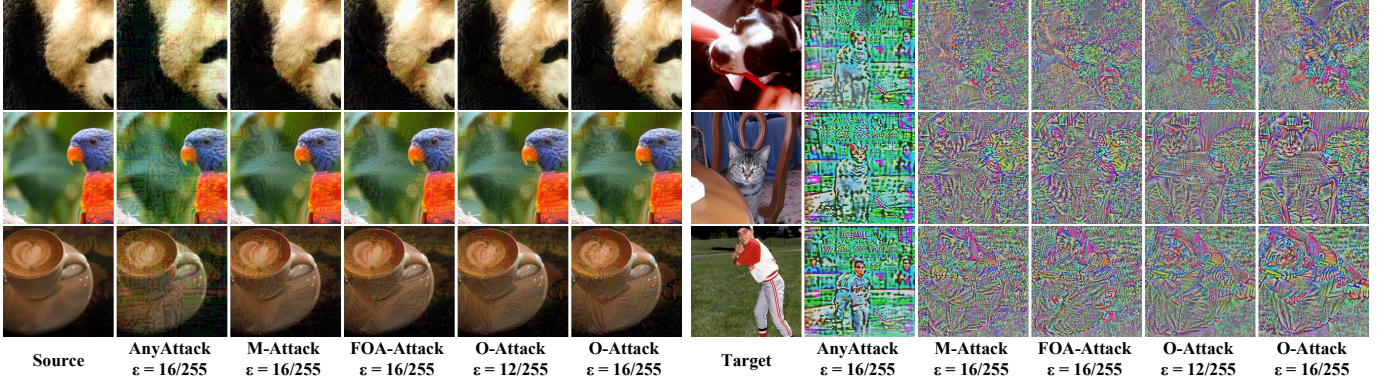


Fig. 8. Qualitative comparison of adversarial examples and their perturbation textures. The left panel shows source images and the adversarial examples generated by different attacks, while the right panel shows target images and the corresponding amplified perturbations. Compared with the baselines, 0-Attack produces cleaner and more coherent target-related semantic structures, including under the smaller $\epsilon = 12/255$ budget.

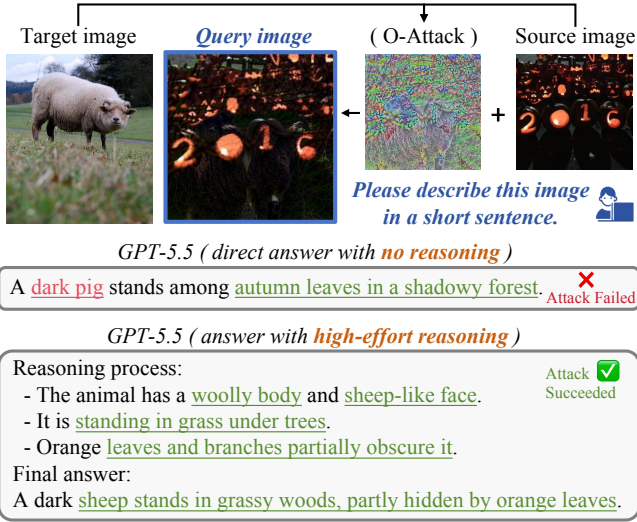


Fig. 9. Qualitative example illustrating the effect of reasoning effort shown in Fig. 5. Given the same adversarial image generated by 0-Attack, the attack on GPT-5.5 fails under direct answering without reasoning but succeeds when high-effort reasoning is enabled, with the model producing a response semantically aligned with the target image.

its perturbation along high-level target-semantic directions rather than merely fitting surrogate-specific low-level noise.

5.4 Ablation Studies

Effect of each component. Table 5 evaluates CSA, PSS, and SCO on six frontier commercial MLLMs. Removing PSS from the full method lowers average ASR from 81.4% to 71.2%, while removing textual guidance from CSA reduces it to 64.2%. Both removals degrade ASR on every victim, showing that the average drops are not driven by a single outlier model. A variant retaining only SCO reaches 59.8%, substantially below the complete framework. These results show that semantic anchoring, progressive sampling, and consensus optimization provide complementary gains.

Effect of surrogate models. Table 6 studies the surrogate CLIP backbones. With the standard three-surrogate set, 0-Attack achieves 81.4% average ASR, and adding CLIP-B/32(l) further raises it to 82.2%. Under the matched four-surrogate setting, this is 13.0 percentage points higher than M-Attack-V2, which reaches 69.2%. Performance decreases gradually to 74.9%, 70.3%, and 60.0% as the ensemble is

reduced to two or one surrogate. Even the single-surrogate variant remains above FOA-Attack with three surrogates (46.2%), indicating that the gains arise primarily from how 0-Attack exploits semantic space.

Effect of crop number K and efficiency. Table 7 varies the number of augmentation views from $K = 3$ to $K = 10$. Average ASR increases overall from 71.7% to 81.4%, showing that additional semantic views strengthen transferability at the cost of longer optimization. Even at $K = 3$, 0-Attack reaches 71.7% ASR and surpasses MPCAttack (69.7%) while using three instead of six surrogate models. At $K = 10$, 0-Attack requires approximately 245 seconds per image, remaining faster than M-Attack-V2 and FOA-Attack while improving their ASRs by 12.2 and 35.2 percentage points, respectively. This provides a favorable effectiveness–efficiency trade-off using only three surrogate models.

Progressive dropout schedule. Table 8 studies how the dropout cap is scheduled in PSS. The progressive cap schedule $0.10 \rightarrow 0.15$ for CLIP-G/14 and $0.05 \rightarrow 0.10$ for CLIP-B/16 and CLIP-B/32 performs best, reaching 81.4% ASR and 0.673 AvgSim. Alternative increasing or decreasing schedules obtain only 72.7%–75.8% ASR. This confirms that transferability depends not only on injecting stochasticity, but also on expanding it from a moderate initial state without making the surrogate distribution excessively unstable.

6 CONCLUSION

In this paper, we revisit transferable adversarial attacks against multimodal large language models and identify that prior methods are fundamentally limited by their reliance on a single final-layer semantic state, which causes overfitting to the surrogate and weak cross-model transfer. Motivated by this, we propose 0-Attack, which exploits the cross-modal high-level semantic space through cross-modal semantic anchoring, progressive semantic space sampling, and semantic consensus optimization. Extensive experiments on 24 MLLMs show that 0-Attack consistently surpasses state-of-the-art baselines, raising the average attack success rate by a large margin on 10 frontier commercial MLLMs and 14 widely used MLLMs, while generalizing across task prompts and perturbation budgets and maintaining visual imperceptibility. We hope our findings encourage further study of the semantic-space vulnerabilities of MLLMs and the development of more robust multimodal systems.

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